

MXET 375 Exam Formula Sheet

Translational Inertia Element (Mass):

$$m\ddot{x}(t) = F_m(t) = \Sigma F$$

Translational Stiffness Element

(Spring):

$$F_k = k[x_2(t) - x_1(t)]$$

Translational Damping Element

(Damper):

$$F_b(t) = b[\dot{x}_2(t) - \dot{x}_1(t)]$$

Rotational Inertia Element:

$$J\ddot{\theta} = T_j(t), \theta_1$$

Rotational Stiffness Element

(Spring):

$$T_k(t) = k[\theta_2(t) - \theta_1(t)]$$

Rotational Damping Element

(Damper):

$$T_b(t) = b[\dot{\theta}(t) - \dot{\theta}_1(t)]$$

Ideal Lever:

$$F_1 L_1 = F_2 L_2$$

Gear Ratio:

$$\frac{r_2}{r_1} = \frac{n_2}{n_1} = N$$

Equating Mesh Velocities:

$$V_{mesh} = r_1 \omega_1 = r_2 \omega_2$$

Equating Input Power = Output
Power:

$$T_1 \omega_1 = T_2 \omega_2$$

$$\frac{T_1}{T_2} = \frac{\omega_2}{\omega_1}$$

$$GR = N = \frac{n_2}{n_1} = \frac{r_2}{r_1} = \frac{D_2}{D_1} = \frac{T_2}{T_1} = \frac{\omega_1}{\omega_2}$$

Pendulum:

$$I\ddot{\theta} = T_c(t) = \Sigma F, \theta_1$$

